Victoria Doyon

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EDUCATION

University of Utah
M.S. Computer Science *Expected: Spring 2029*

Florida Polytechnic University
B.S. Computer Engineering (Machine Intelligence concentration) GPA: 3.97 *Graduated: Spring 2022*

RESEARCH EXPERIENCE

Independent Study, University of Utah
Course Advisor: Dr. Marino *August 2026 - Present*

- Developing a realistic human-feedback user simulator to train LLM agents with feedback that reflects how a real person would react.
- Training the simulator with Text2Grad, grounding its feedback in psychology and neuroscience research, and investigating whether learning like a human leads to human-like behavior.

Senior Capstone: Driving Simulation and Scenario Generation on a Digital City Map [1]
Florida Polytechnic University *August 2021 - Apr 2022*

- Led the design, training, and evaluation of deep learning models for autonomous steering control from front-facing camera input; implemented CNNs based on NVIDIA's end-to-end self-driving model, with dropout-enhanced variants trained on 30,000+ synthetic driving images.
- Benchmarked traditional computer vision against deep learning, selected the best-performing architecture, and deployed it through a real-time Python-MATLAB interface within a Unreal Engine-based simulation for the project showcase.

PUBLISHED WORK

[1] A. Dubs, V. Correa Andrade, M. Ellis, S. Ganley, B. Karaman, and O. Toker, "A Photo-Realistic Simulation and Test Platform for Autonomous Vehicles Research," in Proceedings of the 7th North American International Conference on Industrial Engineering and Operations Management, Orlando, FL, USA, June 11–14, 2022. IEOM Society International. DOI: 10.46254/NA07.20220424

PROFESSIONAL EXPERIENCE

Center for High Performance Computing, University of Utah
Program Assistant (part-time) *July 2026 - Present*

- Building an LLM inference gateway (Open WebUI, LiteLLM, Keycloak) on Kubernetes, standing up the deployment from scratch with no existing deployment pipeline; in development, not yet in use.
- Implementing security controls and guardrails and reviewing university policies to prepare the gateway for One-U RAI review.
- Provided AI-related technical input on Redtail, Utah's new AI factory supercomputer operated by CHPC, drawing on production AI experience from Microsoft.

Microsoft
Software Engineer II *Sep 2024 - July 2026*

- Technical lead for a team of 4+ engineers building a specialized data-retrieval and workflow-automation agent within a hierarchical multi-agent system; owned the sub-agent architecture end to end, drove technical decisions for the team's scope, and contributed to system-level architecture discussions.
- Designed and shipped plugins for an enterprise AI orchestration platform, adding intelligent data retrieval, dynamic summarization, and user-driven data analysis; integrated plugin behavior with orchestrator routing and agent reasoning flows.
- Independently led the design and execution of an evaluation program for an agentic AI system: built and curated evaluation datasets and engineered automated pipelines measuring response quality and accuracy, safety and compliance guardrails, latency and performance, and agent behavior correctness.

- Architected a data-retrieval platform (Cosmos DB, Azure AI Search, C#, Python, Node.js) that replaced a static data model with an automated daily ingestion and indexing pipeline, continuously refreshing structured content for generative-AI workflows.
- Automated large-scale vectorization pipelines (Python, Azure Functions), processing 200 GB of production data in two weeks, eliminating stale embeddings and improving downstream content quality and model grounding.
- Owned Responsible AI and data-safety assessments across engineering teams, applying Microsoft's practices to ensure AI features met safety, security, and performance standards.

Software Engineer *Jun 2022 - Sep 2024*
- Engineered an end-to-end anomaly-detection system (Kusto/KQL, Azure Functions, C#, JavaScript, Azure Anomaly Detector) that cut anomaly-notification noise by 93.3% (about 30 alerts/week to 1 to 2). Built its accessible React KPI dashboard, which streamlined monthly report creation, and the Azure Data Explorer sync that feeds it, with detailed documentation and accessibility testing.
- Delivered an AI-powered content-rewriting tool at the 2024 Microsoft Global Hackathon, using LLMs to generate actionable insights for senior stakeholders and accelerate support-documentation workflows.
- Provided 24/7 support and responded promptly to critical system issues to minimize downtime.
- Strengthened security by adopting federated credentials, Managed Identities, auditing, role refinements, token checks, AME compliance, and restricted SQL access.
- Served as Scrum Master: coordinated team standups, mentored new teammates, and fostered collaboration to resolve bugs and onboard to systems.
- Organized Monthly Service Reviews, compiling data and presenting findings to stakeholders on issues such as high latency and dropped availability.
- Designed an Impact Tracking Template to help users with retrospectives, goal reviews, and quarterly performance evaluations.

HereAfter AI
Engineer *May 2021 - May 2022*
- Designed and shipped a classification system that automatically annotated 15,000 customer files into 45 categories using ML and NLP, replacing fragile rule-based logic with sentence-similarity and deep learning pipelines and meaningfully improving accuracy.
- Built a deep learning evaluation pipeline benchmarking RoBERTa, BERT, and ELECTRA embeddings via HuggingFace Transformers, selecting the optimal architecture for production deployment.
- Analyzed 6,000+ turns and 25 customer-specific Natural Language Understanding annotation sets to improve Human-Computer Interaction quality of the product.
- Implemented a hybrid app enabling clients to record, store, and share personal stories, decreasing content turnaround time by 63%.

TEACHING AND MENTORING

Global Mentorship Initiative
Mentor *Jan 2024 - Apr 2024*
- Completed 14 weekly one-on-one sessions mentoring a university student through a career-readiness program: resume/CV, LinkedIn and professional network, SMART career plan, communication skills, interview preparation, and job search strategy.

Florida Polytechnic University
Student Education Assistant *Sep 2019 - May 2021*
- Tutored lower-level math courses (Precalculus, Calculus 1, Calculus 2) and modeled effective problem-solving techniques.
- Helped students succeed in a math-heavy major by returning to fundamentals and rebuilding missing base knowledge up to the current course.
- Supported students in setting and achieving academic goals.

LEADERSHIP EXPERIENCE

Microsoft Learn Student Ambassadors
Microsoft Learn Student Ambassador *Jul 2020 - May 2022*
- Created a virtual guide that helped 500+ Students Ambassador get acclimated to the program.
- Coordinated and led technological enhancement events to help participants be more marketable.

Microsoft Learn Student Partner *Sep 2019 - Jun 2020*
- Hosted and organized 20+ events; spoke at events with 100+ attendees.
- Created a mentorship program that integrated 100 new Student Partners into the program.

Student Government Association, Florida Polytechnic University
Executive Chief of Staff *May 2021 - May 2022*
- Increased the thoroughness of transcribed notes by 300% and ensured compliance with Sunshine Law.
- Ensured efficient and accurate communication between the Legislative and Executive branches, coordinating 15 fellow SGA members.

Student Organization Committee Chair *May 2020 - Apr 2021*
- Supervised the operations of over 30 student-led clubs and onboarded four new organizations.
- Facilitated learning of the software required for student life operations on campus.

IEEE - HKN Mu Omega
President *Jan 2022 - Apr 2022*
- Planned and ran the chapter's induction ceremony (27 inductees, Apr 2022) and met regularly with the executive board (president, vice president, treasurer, and advisor).

Vice President *Apr 2021 - Dec 2021*
- Recruited 18 members to enhance chapter engagement and collaboration.
- Facilitated clear communication between the board and members, ensuring alignment on goals and initiatives.
- Coordinated interviews for 30+ applicants, streamlining the selection process.

Founding Treasurer *Mar 2020 - Apr 2021*
- Organized the chapter budget and established and managed spending accounts for chapter activities.
- Played a key role in recruiting nine founding members and coordinated ceremonies to welcome new members.

Florida Polytechnic University
Orientation Leader *May 2021 - Aug 2021*
- Facilitated three days of activities (icebreakers, schedule management, Q&A) for 30+ newly admitted students.

SKILLS

AI/ML: RAG systems | Agentic AI | LLM | NLP | Vector Databases | MLOps | Transformers | PyTorch | TensorFlow | Scikit-learn | Computer Vision | OpenAI | Prompt Engineering | Deep Learning

Languages & Frameworks: Python | C#/.NET | Javascript | Typescript | Node.js | React | Next.js | LangChain | Semantic Kernel | AutoGen

Cloud & Infrastructure: Azure Functions | Cosmos DB | Azure AI Search | Infrastructure as Code | CI/CD Pipelines | DevOps | Microservices | Docker | Kubernetes